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LOS Assignment Theory-2

Compare Permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

```
anonymous@anonymous-supreme in ~ via 🍷 v3.14.7
➤ touch sample.txt

anonymous@anonymous-supreme in ~ via 🍷 v3.14.7
➤ mkdir sample_dir

anonymous@anonymous-supreme in ~ via 🍷 v3.14.7
➤ touch sample_dir/inside.txt

anonymous@anonymous-supreme in ~ via 🍷 v3.14.7
[*] ➤ ls -l sample.txt
-rw-r--r-- 0 anonymous 22 Aug 17:38 sample.txt

anonymous@anonymous-supreme in ~ via 🍷 v3.14.7
➤ ls -ld sample_dir
drwxr-xr-x - anonymous 22 Aug 17:39 sample_dir

anonymous@anonymous-supreme in ~ via 🍷 v3.14.7
➤ ls -l sample_dir
-rw-r--r-- 0 anonymous 22 Aug 17:39 inside.txt
```

Final Observation Table

Permission	Regular File – Observation	Directory – Observation
400 (r--)	Can read, cannot modify	Cannot list or enter
200 (-w-)	Cannot read, can modify	Cannot list, enter, or normally create files
100 (--x)	Cannot read; execution allowed by permission bits	Cannot list, but can enter/access if the name is known
500 (r-x)	Can read, cannot modify	Can list and enter, cannot create/delete
600 (rw-)	Can read and modify	Can list, but cannot enter/access contents
700 (rwx)	Can read, modify, and execute permission is allowed	Can list, enter, create, and delete

Questions

What does read (r) permission allow you to do with a regular file?

- It allows you to read or view the contents of the file

What does read (r) permission allow you to do with a directory?

- It allows you to list the files inside the directory.

What does write (w) permission mean for a regular file?

- It allows you to modify or change the content of the file

What does write (w) permission mean for a directory?

- It allows you to create, delete, or rename files inside the directory

What does execute (x) permission mean for a regular file?

- It allows you to run the file as a program or script

Why is execute (x) permission important for accessing a directory?

- It allows you to enter and access the directory

Which permission combination allows the owner to fully access a regular file?

- 700(rwx)

Which permission combination allows the owner to fully access a directory?

- 700(rwx)

Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

- It is because a file contains data, while a directory contains files and folders. So, the permissions have different purposes for each.